

Math 421/510: Homework 5

1. Given $\phi : \mathbb{R} \rightarrow \mathbb{R}$ we define the function $f_{r,\delta} : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f_{r,\delta}(x, y) = \frac{1}{r} \phi\left(\frac{x}{r}\right) \frac{1}{\delta} \phi\left(\frac{y}{\delta}\right).$$

- (a) Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that

$$\text{supp } \phi \subseteq [-2, 2], \quad \phi(x) = 1 \text{ for } x \in [-1, 1].$$

Let $r, \delta > 0$, assume r is large and δ is small, and Note that $f_{r,\delta}$ is essentially supported/concentrated in a horizontal rectangle of size $\sim r \times \delta$. Compute $\widehat{f_{r,\delta}}$ and determine where it is concentrated in frequency space.

- (b) Choose an explicit function of ϕ satisfying the assumptions above (Hint: if you choose a trapezoidal bump function, it can be written as a convolution of two characteristic functions on an interval) and compute $\widehat{\phi}$ explicitly. Use this example to illustrate the concentration properties found in part (a) and check Plancherel's identity for $f_{r,\delta}$.
- (c) Repeat (a) and (b) for the case $\phi(x) = e^{-x^2}$. Compute the Fourier transform explicitly and verify Plancherel in this case.
- (d) Do the conclusions become stronger if ϕ is smoother? Justify your answer.

Solution.

- (a) We compute $\widehat{f_{r,\delta}}$. We justify the use of Fubini Tonelli since $f_{r,\delta}e^{-2\pi i \xi \cdot x}$ is compactly supported and bounded so it is in L^1 .

$$\begin{aligned} \widehat{f_{r,\delta}}(\xi_x, \xi_y) &= \int_{\mathbb{R}^2} f_{r,\delta}(x, y) e^{-2\pi i x \xi_x} e^{-2\pi i y \xi_y} \\ &= \int_{\mathbb{R}^2} \frac{1}{r} \phi(x/r) \frac{1}{\delta} \phi(y/\delta) e^{-2\pi i x \xi_x} e^{-2\pi i y \xi_y} \\ &= \int_{\mathbb{R}} \frac{1}{r} \phi(x/r) e^{-2\pi i x \xi_x} dx \int_{\mathbb{R}} \frac{1}{\delta} \phi(y/\delta) e^{-2\pi i y \xi_y} dy \text{ by Fubini-Tonelli} \\ &= \int_{\mathbb{R}} \phi(u) e^{-2\pi i \xi_x r u} du \int_{\mathbb{R}} \phi(v) e^{-2\pi i \delta v \xi_y} dv \\ &= \widehat{\phi}(r \xi_x) \widehat{\phi}(\delta \xi_y). \end{aligned}$$

Since ϕ is in L^1 , we have that $\widehat{\phi} \in C_0$. Therefore, for any $\varepsilon > 0$ there exists some interval $[-M, M]$ such that $\int_{\mathbb{R} \setminus [-M, M]} |\widehat{\phi}| < \varepsilon$.

But then, $\widehat{\phi}(r \xi_x)$ is concentrated in the interval $[-M/r, M/r]$, and $\widehat{\phi}(\delta \xi_y)$ is concentrated in $[-\delta M/\delta, \delta M/\delta]$. If r is large and δ is small, this means that the rectangle $[(-rM, -\delta M), (rM, \delta M)]$ in which $\widehat{f_{r,\delta}}$ is concentrated has a small dimension in the ξ_x frequency axis and a short dimension in the ξ_y frequency axis. Thus, it is concentrated close to the ξ_x axis in frequency space.

- (b) We consider $g = \chi_{[-1,0]}$, and $h = \chi_{[-2,1]}$. We compute $g * h$ to show that it has the desired properties.

$$\begin{aligned} g * h(x) &= \int_{\mathbb{R}} g(x-y)h(y)dy \\ &= \int_{-2}^1 \chi_{[-1,0]}(x-y)dy \\ &= \int_{-2}^1 \chi_{[-1-x,-x]}(-y)dy. \end{aligned}$$

If $|x| > 2$, then $-1-x > 1$ or $-x < -2$, which will mean $\chi_{[-1-x,-x]}$ is disjoint with the domain of integration, so we have $g * h(x) = 0$. Now, suppose that $-2 \leq x \leq -1$. Then,

$$g * h(x) = \int_{-2}^1 \chi_{[-1-x,-x]}(-y)dy = \int_{-1-x}^1 dy = 1 + 1 - x \Rightarrow 0 \leq g * h(x) \leq 1.$$

Suppose $-1 \leq x \leq 1$. Then,

$$g * h(x) = \int_{-2}^1 \chi_{[-1-x,-x]}(-y)dy = \int_{-1-x}^{-x} dy = -x + 1 + x = 1 \Rightarrow g * h(x) = 1.$$

Finally, suppose $1 \leq x \leq 2$. Then,

$$g * h(x) = \int_{-2}^1 \chi_{[-1-x,-x]}(-y)dy = \int_{-2}^{-x} dy = -x + 2 \Rightarrow 0 \leq g * h(x) \leq 1.$$

Since the convolution of L^1 functions is continuous, we have that $g * h$ is continuous, $g * h(x) = 1$ for $x \in [-1, 1]$ (as desired), and $\text{supp } \phi \subseteq [-2, 2]$.

Now, if $\phi = g * h$, then $\hat{\phi} = \hat{g}\hat{h}$. We have that:

$$\begin{aligned} \hat{g} &= \frac{1}{-2\pi i \xi} (1 - e^{2\pi i \xi}) = \frac{1}{\pi \xi} e^{\pi i \xi} \sin(\pi \xi) \\ \hat{h} &= \frac{1}{-2\pi i \xi} (e^{-2\pi i \xi} - e^{4\pi i \xi}) = \frac{1}{\pi \xi} e^{-\pi i \xi} \sin(3\pi \xi). \end{aligned}$$

$$\text{So, } \hat{g}\hat{h} = \frac{1}{\pi^2 \xi^2} \sin(\pi \xi) \sin(3\pi \xi)$$

Then, we have that

$$\widehat{f_{r,\delta}}(\xi_x, \xi_y) = \frac{1}{\pi^4 r^2 \delta^2 \xi_x^2 \xi_y^2} \sin(r 3\pi \xi_x) \sin(r \pi \xi_x) \sin(\delta \pi \xi_y) \sin(\delta 3\pi \xi_y).$$

Close to the origin, we have that $\sin(\xi) \approx \xi$, so then $\frac{1}{\pi^2 \xi^2} \sin(\pi \xi) \sin(3\pi \xi) \approx 3$. When $\xi > 1$ however, the function quickly drops off to zero. Here is the function in Desmos:

How "wide" this bump is, and how close it is to the ξ_x and ξ_y axes, is then determined by these scaling factors δ, r in the respective dimensions. Thus, the bump will be situated in the rectangle $[(-1/r, -1/\delta), (1/r, 1/\delta)]$.

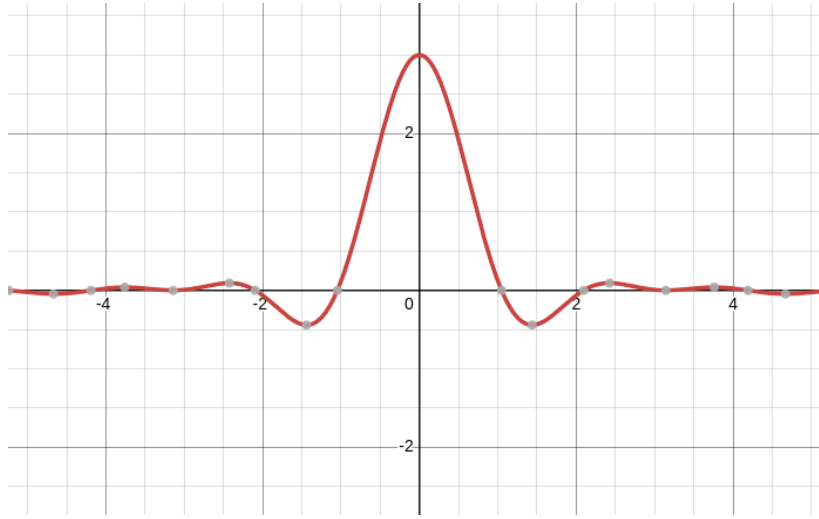


Figure 1: $\frac{1}{\pi^2 x^2} \sin(\pi x) \sin(3\pi x)$

Finally, we check Plancherel's identity. Since f and g are linear functions, their convolution is a trapezoid. In this case, we can easily see that the area of the trapezoid is $\int_{-2}^2 \phi(x) = 1/2 + 2 + 1/2 = 4$. The area under the squared trapezoid, is then $\int_{-2}^2 |\phi(x)|^2 = \frac{1}{3} + 2 + \frac{1}{3} = 8/3$. Then,

$$\begin{aligned} \int_{\mathbb{R}^2} |f_{r,\delta}|^2 &= \int_{\mathbb{R}} \frac{1}{r^2} |\phi(x/r)|^2 dx \int_{\mathbb{R}} \frac{1}{\delta^2} |\phi(y/\delta)|^2 dy \\ &= \frac{1}{r\delta} \int_{\mathbb{R}} |\phi(x)|^2 \int_{\mathbb{R}} |\phi(y)|^2 dy = \frac{64}{9r\delta}. \end{aligned}$$

I'm afraid I will have to take the L on the integral in the frequency domain...I'm not sure how to tackle it.

- (c) By Theorem 5.5 in class, we have that $\mathcal{F}(\phi) = \mathcal{F}(e^{-x^2}) = \frac{1}{\sqrt{\pi}} e^{-\pi^2 \xi^2}$, since here $a = 1/\pi$. Thus,

$$\widehat{f_{r,\delta}}(\xi_x, \xi_y) = \frac{1}{\pi} e^{-\pi^2((r\xi_x)^2 + (\delta\xi_y)^2)}.$$

We use Fubini freely. Now, we verify Plancherel.

$$\int_{\mathbb{R}^2} |f_{r,\delta}|^2 = \frac{1}{r^2 \delta^2} \int_{\mathbb{R}} e^{-2x^2/r^2} dx \int_{\mathbb{R}} e^{-2y^2/\delta^2} dy = \frac{1}{r^2 \delta^2} \left(\sqrt{\frac{r^2}{2\pi}} \right) \left(\sqrt{\frac{\delta^2}{2\pi}} \right) = \frac{1}{r\delta} \frac{1}{2\pi}.$$

Next, we compute:

$$\int_{\mathbb{R}^2} |\widehat{f_{r,\delta}}|^2 = \int_{\mathbb{R}} \frac{1}{\pi} e^{-2\pi^2 r^2 \xi_x^2} d\xi_x \int_{\mathbb{R}} \frac{1}{\pi} e^{-2\pi^2 \delta^2 \xi_y^2} d\xi_y = \frac{1}{\pi} \left(\sqrt{\frac{\pi}{2r^2}} \right) \left(\sqrt{\frac{\pi}{2\delta^2}} \right) = \frac{1}{r\delta} \frac{1}{2\pi}.$$

Thus, we have verified Plancherel's identity.

- (d) If ϕ becomes smoother, then by theorem 5.10 5) $\hat{\phi}$ decays at least as quickly as the power of the highest degree continuous derivative. We can say more precisely how the frequency spectrum will decay. We don't just know that it eventually will (which is what the Riemann-Lebesgue lemma means, in having $\mathcal{F}(L^1) \subset C_0$).

2. On $\mathbb{R}$, let

$$f(x) = \begin{cases} 2x + 1 & -\frac{1}{2} \leq x \leq 0, \\ 1 - 2x & 0 \leq x \leq \frac{1}{2}, \\ 0 & |x| > \frac{1}{2}. \end{cases}$$

- (a) Show that

$$\hat{f}(\xi) = \frac{1 - \cos(\pi\xi)}{\pi^2\xi^2}.$$

- (b) Briefly explain why this computation is consistent with the principle quantified by the relations $\widehat{g^{(n)}}(\xi) = (2\pi i\xi)^n \hat{g}(\xi)$ and $(-2\pi ix)^n g(x) = (\hat{g})^{(n)}(\xi)$ that spatial decay of a function is reflected in the smoothness of its Fourier transform (and vice versa).
- (c) Use your computation from part (a) to show:

i. $\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8};$

ii. $\sum_{n=0}^{\infty} \frac{1}{(2n+1)^4} = \frac{\pi^4}{96}.$

- (a) First, note that $f(x) = \chi_{[-1/2, 1/2]} + 2x\chi_{[-1/2, 0]} - 2x\chi_{[0, 1/2]}$

We compute:

$$\begin{aligned} \hat{f}(\xi) &= \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx \\ &= \int_{-1/2}^{1/2} e^{-2\pi i x \xi} dx + \int_{-1/2}^0 2x e^{-2\pi i x \xi} dx - \int_0^{1/2} 2x e^{-2\pi i x \xi} dx. \end{aligned}$$

The first integral is just $\left. \frac{1}{-2\pi i \xi} e^{-2\pi i x \xi} \right|_{-1/2}^{1/2} = \frac{1}{-2\pi i \xi} (e^{-\pi i \xi} - e^{\pi i \xi}) = \frac{1}{i\pi \xi} \sin(\pi \xi)$. The next two integrals, by integration by parts, is

$$\begin{aligned} \int_a^b 2x e^{-2\pi i x \xi} dx &= \left. \frac{2x}{-2\pi i \xi} e^{-2\pi i x \xi} \right|_a^b - \int_a^b \frac{1}{-\pi i \xi} e^{-2\pi i x \xi} dx \\ &= \frac{1}{-\pi i \xi} (b e^{-2\pi i b \xi} - a e^{-2\pi i a \xi}) - \frac{1}{-2\pi^2 \xi^2} e^{-2\pi i x \xi} \Big|_a^b. \end{aligned}$$

Plugging this in with the respective bounds $-1/2, 0$ and $0, 1/2$, we get

$$\begin{aligned}\hat{f}(\xi) &= \frac{1}{\pi i \xi} \sin(\pi \xi) + \left(\frac{1}{-2\pi i \xi} e^{\pi i \xi} + \frac{1}{2\pi^2 \xi^2} (1 - e^{\pi i \xi}) \right) - \left(-\frac{1}{2\pi i \xi} e^{-\pi i \xi} + \frac{1}{2\pi^2 \xi^2} (e^{-\pi i \xi} - 1) \right) \\ &= \frac{1}{\pi i \xi} \sin(\pi \xi) - \frac{1}{\pi i \xi} \sin(\pi \xi) + \frac{1}{\pi^2 \xi^2} \left(1 - \frac{1}{2} e^{\pi i \xi} - \frac{1}{2} e^{-\pi i \xi} \right) \\ &= \frac{1}{\pi^2 \xi^2} (1 - \cos(\pi \xi)),\end{aligned}$$

as desired. Also, we note that since the fourier transform is uniformly continuous when f is in L^1 (as is the case here), we compute $\hat{f}(0)$ by a double application of l'Hopital's rule:

$$\hat{f}(0) = \lim_{\xi \rightarrow 0} \hat{f}(\xi) = \frac{\lim_{\xi \rightarrow 0} (1 - \cos(\pi \xi))}{\lim_{\xi \rightarrow 0} \pi^2 \xi^2} = \frac{\lim_{\xi \rightarrow 0} \pi \sin(\pi \xi)}{\lim_{\xi \rightarrow 0} 2\pi^2 \xi} = \frac{\lim_{\xi \rightarrow 0} \pi^2 \cos(\pi \xi)}{\lim_{\xi \rightarrow 0} 2\pi^2} = \frac{1}{2}.$$

- (b) Since f is not particularly smooth (it doesn't have a continuous derivative... it is continuous, but the only guarantee that gives us is that the fourier transform is in C_0), we can't say much about it. However, f has immediate spacial decay. Indeed, it is in C_c , when we see that $\text{supp } f = [-1, 1]$. Therefore, it checks out that $\hat{f}$ is infinitely differentiable (in C^∞).
- (c) i. First, we recognize that f is continuous at 0, since $2(0) + 1 = 1 - 2(0) = 1$, and since $|\hat{f}(k)| \leq \frac{2}{\pi^2 k^2}$ for each $k > 0$, we have that

$$\sum_{k \in \mathbb{Z}} |\hat{f}(k)| \leq \frac{1}{2} + 2 \cdot \frac{2}{\pi^2 \xi^2} \sum_{n=1}^{\infty} \frac{1}{k^2} < \infty.$$

Therefore, we can apply the Poisson summation formula, since it's conditions are satisfied. But since $f(x) = 0$ for $|x| > 1/2$, we have $\sum_{m \in \mathbb{Z}} f(m) = f(0)$. Also, for $k \in \mathbb{Z}$, we have that $\frac{1 - \cos(\pi k)}{\pi^2 k^2} = 0$ when k is non-zero even and $\frac{2}{\pi^2 k^2}$ when k is odd. Thus,

$$\begin{aligned}\sum_{k \in \mathbb{Z}} \hat{f}(k) &= \hat{f}(0) + 2 \cdot \sum_{k=1}^{\infty} \hat{f}(k) \text{ since } \hat{f} \text{ is an even function} \\ &= \frac{1}{2} + 2 \cdot \sum_{n=0}^{\infty} \hat{f}(2n+1) \text{ since } \hat{f}(k) = 0 \text{ when } k \text{ even} \\ &= \frac{1}{2} + 2 \frac{1}{\pi^2} \sum_{n=0}^{\infty} \frac{2}{(2n+1)^2}.\end{aligned}$$

Applying the Poisson integral formula,

$$\begin{aligned}
\sum_{m \in \mathbb{Z}} f(m) &= \sum_{k \in \mathbb{Z}} \hat{f}(k) \\
1 &= \frac{1}{2} + 2 \frac{1}{\pi^2} \sum_{n=0}^{\infty} \frac{2}{(2n+1)^2} \\
\frac{1}{2} &= \frac{4}{\pi^2} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} \\
\Rightarrow \frac{\pi^2}{8} &= \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2},
\end{aligned}$$

as desired.

ii. First, we see that

$$\begin{aligned}
\|Pf\|_{L^2(\mathbb{T})}^2 &= \int_{\mathbb{R}} |f|^2 = \int_{-1/2}^0 (2x+1)^2 + \int_0^{1/2} (1-2x)^2 \\
&= \frac{1}{2} \int_0^1 u^2 du - \frac{1}{2} \int_1^0 u^2 du \\
&= \frac{1}{6} + \frac{1}{6} = \frac{1}{3}.
\end{aligned}$$

By Theorem 5.33, $\widehat{Pf}(k) = \hat{f}(k)$, and by the Parseval identity, we have $\sum_{k \in \mathbb{Z}} |\hat{f}(k)|^2 = \|Pf\|_{L^2(\mathbb{T})}^2$. Arguing similarly about the evenness of $\hat{f}$ and how it is zero on the non-zero even integers, we compute this sum:

$$\begin{aligned}
\sum_{k \in \mathbb{Z}} |\hat{f}(k)|^2 &= \frac{1}{2^2} + 2 \cdot \sum_{n=0}^{\infty} |\hat{f}(2n+1)|^2 \\
&= \frac{1}{2^2} + 2 \cdot \sum_{n=0}^{\infty} \frac{4}{\pi^4 (2n+1)^4} \\
&= \frac{1}{4} + \frac{8}{\pi^4} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^4}.
\end{aligned}$$

With Parseval, we get

$$\begin{aligned}
\frac{1}{3} &= \frac{1}{4} + \frac{8}{\pi^4} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^4} \\
\frac{1}{12} &= \frac{8}{\pi^4} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^4} \\
\frac{\pi^4}{96} &= \sum_{n=0}^{\infty} \frac{1}{(2n+1)^4},
\end{aligned}$$

as desired.